

Online Teaching Assistant

Team no. : 21

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Introduction

The Purpose of the project is to improve the effectiveness of Online education by Creating single Integrated platform for both students and teachers.

In many existing system students face delays in doubt resolution and difficulty in accessing learning materials , while Teachers spend considerable time managing courses , resources and student's information accross multiple tools.Our Project aims to address these issues.

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Addressing Critical Pain Points in EdTech

Current educational tools often create more friction than they solve. Students face delays in getting crucial information, while teachers struggle with administrative burdens.



Student Challenges

- Slow doubt resolution
- Dispersed course materials
- Lack of personalized support



Teacher Challenges

- Tedious course creation
- Fragmented resource management
- Inefficient student tracking

The Problem with Current Solutions

Existing tools (LMS, email, chatbots) are isolated and provide a **poor user experience**, leading to inefficiency and frustration.

Our Motivation

To significantly **reduce teacher overhead** and **accelerate student learning cycles** through a unified, intelligent platform.

User Journeys: Simplifying Education for All

The OTA platform is designed with clear, intuitive pathways for every user, from prospective students to dedicated educators.

1

Student Flow

- Sign up & explore courses
- Enroll & access materials
- Engage AI for doubt resolution

2

Teacher Flow

- Register & create courses
- Upload PDFs & resources
- Monitor enrolled students

Core Features: Empowering Learning & Teaching

The Online Teaching Assistant is built upon a robust set of features designed to enhance both the learning and teaching experience.



Role-Based Authentication

Secure login and access controls distinguishing between student and teacher functionalities.



Dynamic Course Management

Interactive course cards and detailed pages with integrated PDF attachments for comprehensive learning.



Enrollment & Analytics

Track student enrollments, monitor course participation, and view basic performance metrics.



Integrated AI Chat

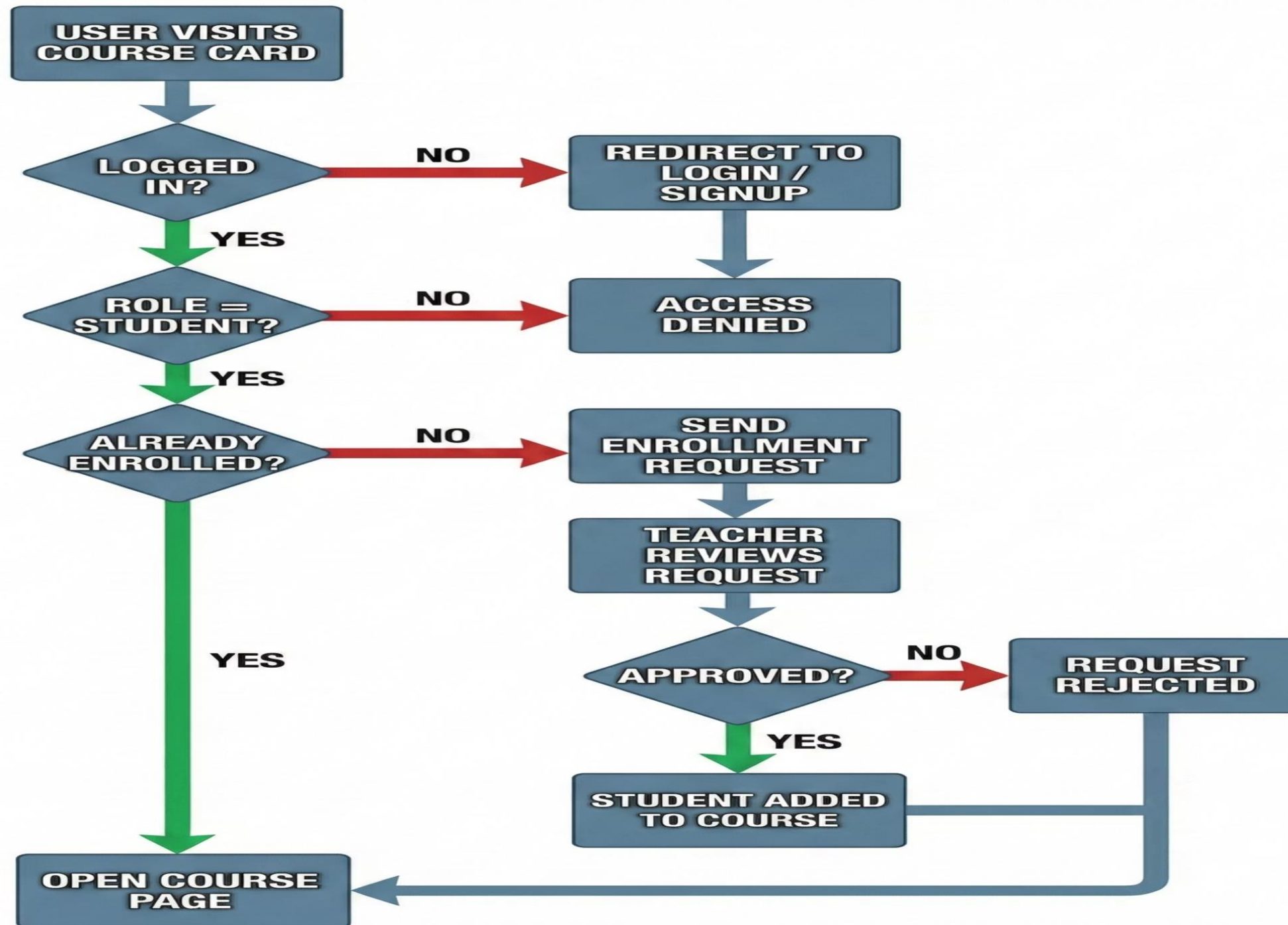
Persistent AI chat history for Q&A and instant content summarization, personalized for each user.



Responsive UI

A modern, adaptable interface built with Material UI and Tailwind CSS for optimal viewing on any device.

Flowchart



Teacher coursecard Flowchart

FRAME 1: Teacher Portal - Create Course UI

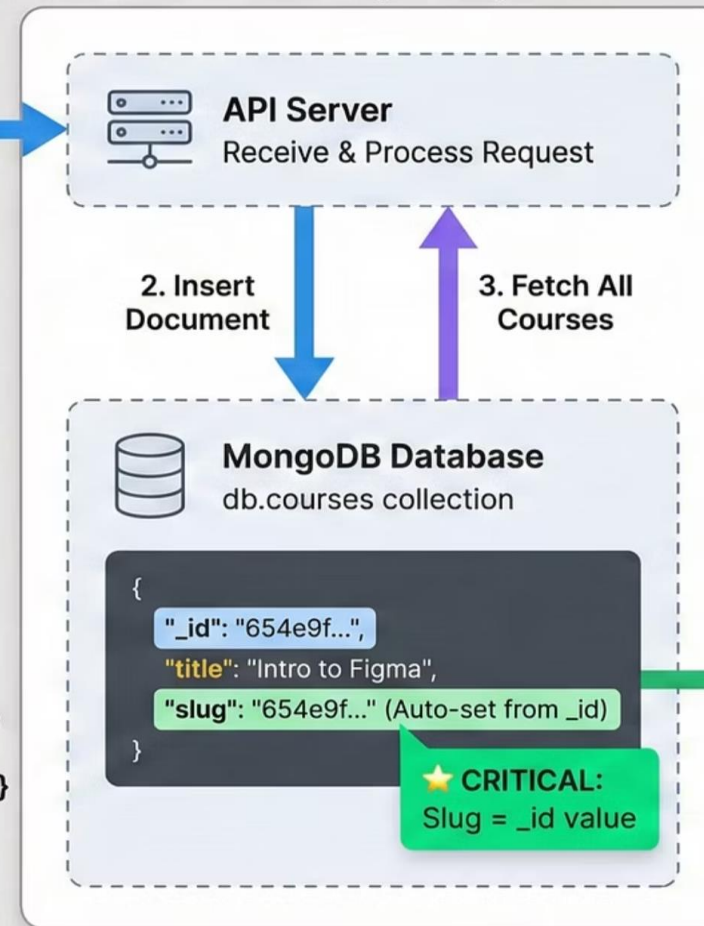
Create New Course

Course Title

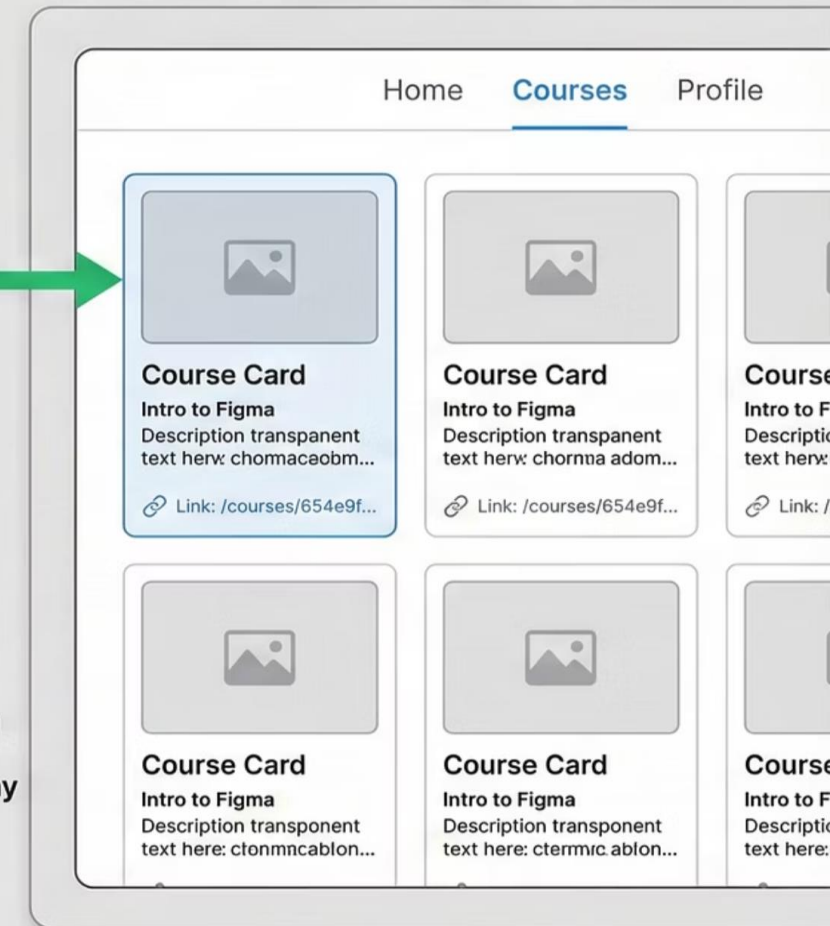
Description

Publish Course

FRAME 2: Backend & MongoDB Logic

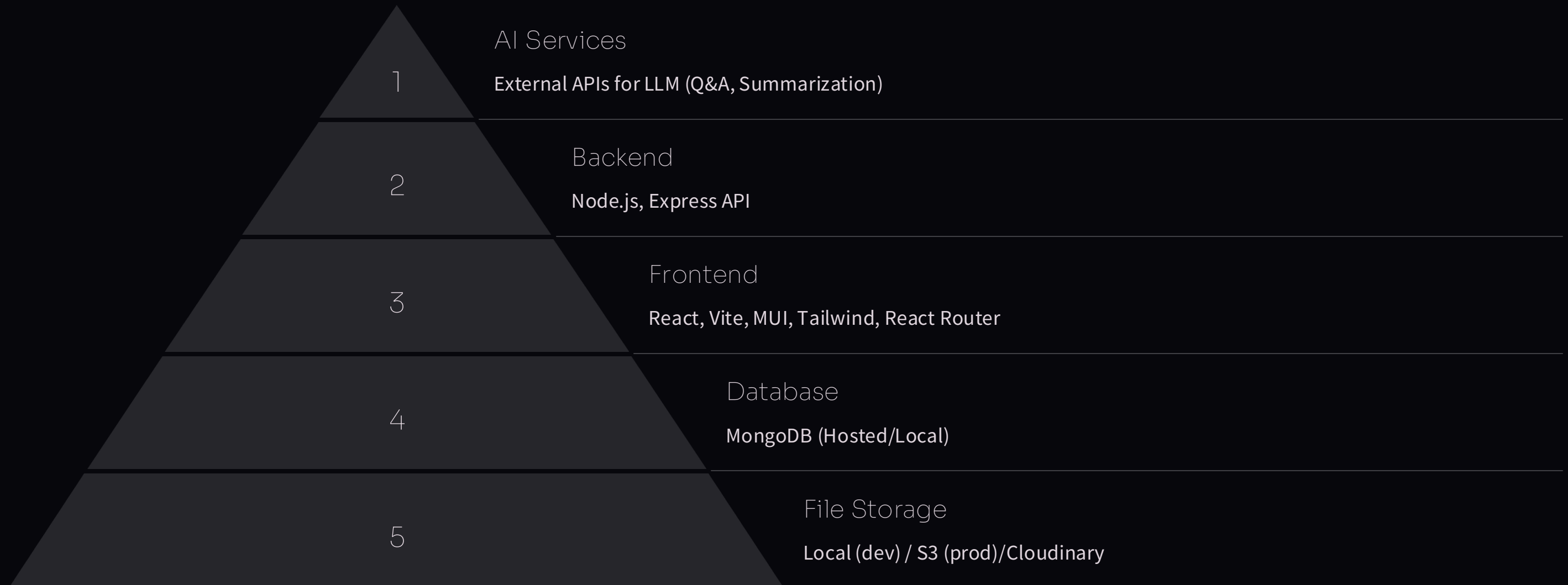


FRAME 3: Student Website - Courses Tab UI



Architectural Blueprint: A Scalable & Robust Foundation

The Online Teaching Assistant is built on a modern, flexible architecture designed for performance and future expansion.



Data Model: The Backbone of the OTA Platform

A well-structured MongoDB schema is crucial for efficiently managing user data, course content, and AI interactions.

User Collection

- **name, email (unique), password (hashed)**
- **role:** student | teacher | admin
- **profileDetails:** avatar, bio, department, university
- **enrolledCourses[]** (student → course references)
- **createdCourses[]** (teacher → course references)
- **lastLoginAt** (tracking user activity)
- **accountStatus:** active / suspended
- **timestamps:** createdAt, updatedAt

Course Collection

- **title, description, subjectCategory**
- **teacherId** (reference to User)
- **students[]** (array of User references)
- **courseStatus:** draft / published / archived
- **files[]** (PDF metadata):
 - fileName
 - fileUrl (Cloud storage)
 - fileSize
 - uploadDate

AIChat Collection

- **userId** (reference to User)
- **courseId** (optional – context-based chat)
- **question**
- **answer**
- **aiModelUsed** (e.g., Gemini / Groq)
- **responseTime**
- **timestamps**

Future Enhancements

Schemas for **Quizzes** and **Assignments** to enrich assessment capabilities.

Security Measures: Protecting Your Data

The OTA platform is built with a strong focus on security, safeguarding user data and ensuring a trustworthy learning environment.



Secure Authentication

Robust user authentication with hashed passwords and JSON Web Tokens (JWTs) ensures only authorized individuals access the platform.



Data Encryption

All sensitive data is encrypted, both in transit using SSL/TLS protocols and at rest within our MongoDB database, protecting it from unauthorized access.



Role-Based Access

Granular role-based access control strictly defines permissions for students and teachers, preventing unauthorized access to specific course materials or administrative functions.



Vulnerability Protection

Comprehensive input validation and sanitization techniques are implemented to defend against common web vulnerabilities like SQL injection and Cross-Site Scripting (XSS).

Key Backend APIs: Facilitating Interaction(1)

Our API endpoints enable seamless communication between the frontend and backend, powering all user and administrative actions.

User Authentication

POST /api/auth/signup - Create new user, return JWT.

POST /api/auth/login - Authenticate user, return JWT + user data.

Course Management

GET /api/courses - List all courses (optional authentication).

POST /api/courses - Create a new course (teacher only, authenticated).

POST /api/courses/:id/upload - Upload PDFs for a course (teacher only).

Student Enrollment

POST /api/courses/:id/enroll - Enroll a student in a specific course.

All teacher-specific endpoints are secured with JWT middleware and role-based checks, ensuring data integrity and security.

Key Backend APIs: Facilitating Interaction(2)

AI Interaction

`POST /api/ai/chat` - Send a question to the AI, persist interaction to `AIChat`.

User Profile and role management

- `GET /api/users/me`
→ Fetch logged-in user profile using JWT.
- `GET /api/users/:id`
→ Admin/teacher-level user lookup (optional).

Teacher Dashboard

- `GET /api/teacher/courses`
→ Fetch all courses created by the teacher.
- `GET /api/teacher/students/:courseId`
→ View enrolled students for a course.
- `GET /api/teacher/analytics/:courseId`
→ Basic insights (student count, file usage, AI queries).

AI/ML Integration: Smart Learning, Smart Teaching

The heart of our Online Teaching Assistant lies in its intelligent AI/ML capabilities, designed to provide instant support a

→ Doubt Solver

User queries are routed to an external Large Language Model (LLM) or specialized tool for accurate, context-aware answers.

→ Persistent Chat History

All AI interactions are stored per user, allowing for review, personalized learning paths, and future AI model training.

→ Optional: PDF Summarization

AI can analyze uploaded PDF documents and automatically generate concise notes or summaries, saving study time.

→ Future Scope: TTS & Avatars

Integration of Text-to-Speech (TTS) for audio replies and AI-driven video avatars for more engaging interactions.



Enhanced AI Capabilities: Intelligent Learning Support

Building on our core AI integration, the OTA platform offers advanced context-aware assistance and comprehensive summarization tools to maximize learning efficiency.



Context-Aware Chatbot

Our AI chatbot understands the current content (document, video, or lesson) to provide precise, relevant answers, significantly enhancing comprehension and interaction.



On-Demand Document Summaries

Instantly generate concise summaries of uploaded PDF documents at any point, highlighting key concepts and saving valuable study and preparation time.



Smart Video Insights

Quickly get key takeaways from uploaded video lectures, including automatically generated transcripts and summary points, streamlining video-based learning.

Student Analytics & Performance Monitoring

The OTA platform meticulously collects and analyzes student performance data, providing valuable insights for both learners and educators to optimize the educational journey.



Comprehensive Data Collection

We track engagement metrics, quiz results, assignment completion, AI interaction patterns, and time spent on course materials to build a holistic student profile.



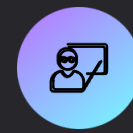
Intuitive Performance Dashboards

Students and teachers gain access to visual dashboards that highlight progress, identify strengths, and pinpoint areas requiring more attention, promoting data-driven learning.



Proactive Learning Gap Identification

AI-powered algorithms analyze performance trends to proactively identify potential learning gaps, allowing for timely intervention and personalized remedial recommendations.



Empowering Teacher Insights

Educators receive detailed reports on class and individual student performance, enabling them to tailor teaching strategies and provide targeted support where it's needed most.

Competitive Advantages: What Sets Us Apart

The Online Teaching Assistant platform distinguishes itself through key features designed for a superior educational experience.



AI-Powered Learning

Our integrated AI offers instant answers, content summarization, and personalized assistance, making learning more efficient and engaging.



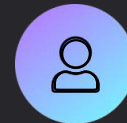
Robust Security

We prioritize data protection with advanced encryption, secure authentication, and strict access controls, ensuring a safe environment for all users.



Scalable Architecture

Built on a modern tech stack, the platform is designed for high performance, reliability, and easy expansion to accommodate future growth and features.



Intuitive Design

A clean, responsive, and user-friendly interface ensures seamless navigation and interaction for both students and teachers on any device.

These core strengths ensure the OTA platform delivers unmatched value in the evolving EdTech landscape.

Pricing Model: Value for Every User

Our flexible pricing structure ensures that the Online Teaching Assistant is accessible and valuable for diverse user needs, from individual students to large institutions.



Free Tier

Basic access, limited AI features.
Perfect for new users.



Premium Plan

Full AI capabilities, unlimited courses, priority support. Ideal for educators and power users.



Institutional License

Custom features, bulk user management, dedicated support. Designed for schools and universities.

Future Advancements: Expanding OTA's Capabilities

The Online Teaching Assistant platform has a roadmap for continuous innovation, focusing on potential features that will further empower learners and educators.



Adaptive Learning Paths & Analytics

Implement AI-driven adaptive learning paths and provide educators with advanced analytics for deeper insights into student performance.



Dedicated Mobile Application

Develop a dedicated mobile app for seamless learning on-the-go, ensuring accessibility across all devices and operating systems.



Advanced Assessment Tools

Integrate sophisticated assessment tools, including AI-powered grading and diverse question types, for comprehensive evaluation.



Real-time Collaboration & Video

Introduce real-time collaboration features and integrated video conferencing to foster interactive group learning experiences.



Gamification & Multilingual Support

Enhance engagement through gamification elements and expand global reach with comprehensive multilingual support.

We are dedicated to evolving OTA with these advancements, continuously adding valuable features to maintain its leadership in the global EdTech landscape.

Conclusion

The Online Teaching Assistant (OTA) platform represents a significant leap forward in educational technology, designed to meet the evolving demands of both educators and learners.



Empowering Learning

AI-driven tools offer personalized support, instant answers, and enhanced content engagement for a more effective educational experience.



Secure & Reliable

Built with robust security measures and a scalable architecture, OTA ensures a safe, private, and high-performing environment for all users.



Future-Proof Innovation

With a clear roadmap for continuous advancement, OTA is committed to integrating cutting-edge features that will define the future of online education.

We are confident that OTA will transform the digital classroom, fostering an intelligent, interactive, and inclusive learning ecosystem.

References & Resources

Key sources and foundational research that informed the development of the Online Teaching Assistant platform.

- Russell, S. J., & Norvig, P. (2010). *Artificial Intelligence: A Modern Approach* (3rd ed.). Pearson Education.
- <https://cloudinary.com/> , <https://webllm.mlc.ai/> , <https://www.mongodb.com/> website.
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- OpenAI. (2023). *GPT-4 Technical Report*.
- EDUCAUSE Learning Initiative. (2022). *7 Things You Should Know About AI in Teaching and Learning*.

This presentation integrates insights from leading research in artificial intelligence, educational technology, and cybersecurity practices to ensure a robust and effective learning environment.

Acknowledgement

We extend our deepest gratitude to Dr. Munmun Bhattacharya for her invaluable guidance, unwavering support, and profound expertise throughout the development of the Online Teaching Assistant (OTA) platform. Her mentorship was instrumental in shaping this project, and her insights were a constant source of inspiration. This project would not have been possible without her dedicated leadership.

Thank you